Electronics 101 — from the dev board to the class board

Lecture 2 • Fri 18 Sep 2026 • 14:20–16:20 • R311 IAMS

Six ideas on our schematic (the circuit drawing) • KiCad's six operations (KiCad — the free circuit-design program) • place your missing parts • route (draw the copper tracks in) your zone • your first pull request (PR) into the class board

Where we are

- Every phone talks to its ESP32 (E2 merged into the shared project). Every student has a block and a `SPEC.md` with a number.
- The **full schematic** exists (PDF on the site). Your **gapped sheet** is missing 3–4 parts. The **PCB** (the physical board layout) has every footprint (the copper pattern a part is soldered onto) pre-placed and your rule area (the outlined region that is yours to route) `ZONE_B<N>` with your name on it.
- **JLC (JLCPCB, the board factory) assembles everything.** You solder nothing. The dev board stays the fallback.
- **Order: Mon 28 Sep.** Your zone must be merged before it; hand-in dates are agreed in class.

The board, v0.6 — one block per student

Block	What is in your zone	Parts
B1	ADS8688 + 8 × (1 kΩ, 1 nF, BAV99) → AI1–8, ±10 V 16-bit	≈ 32
B2	USB-C, jack, ORing, polyfuses, TVS, 2 × 2 W isolated → ±12 V, 78L05, AMS1117	≈ 20
B3	DAC8563 + OPA2192 ±10 V + 2 × (49.9 Ω + clamp) → AO1, AO2	≈ 15
B4	4 relays via ULN2003, 2 × 6N137, terminals, LEDs	≈ 24
B5	74AHCT541 (8 × TTL), 74HCT125, 74LVC1T45 (TRIG), TCXO (DNP — not fitted)	≈ 15

Base (instructor): socket, buses, LED, Qwiic, header, star point, front panel 3 × 4 SMA at 20 mm.

(a) Rails, decoupling, ground — B2

- 5 V in (USB-C **or** jack, SS34 ORing) → **+5V_RAW** → AMS1117 → **+3V3**
- 2 × B0512S-2WR3 isolated modules → **+12 V / AGND / −12 V** → 78L05 → **+5VA**
- **Rule:** anything not voltage-sensitive runs from raw 5 V. ±12 V and +5VA feed only op-amps, ADC, DAC.
- **100 nF at every supply pin, 10 µF per rail per zone.** Distance beats value.
- **Two grounds, one star point.** GND (power, logic; layer 2 plane) and AGND (signals) meet once, at the B1/base boundary.

(b) What ±10 V and 16 bits buy — B1 / B3

- 20 V / 65,536 = **0.3 mV per LSB**
- Every other choice on the board exists to keep noise below that: the star point, AGND copper under the ADC, no digital under the inputs, SPI entering from one side.
- Input network **1 kΩ → 1 nF → BAV99 to ±12 V**: current limit, gentle low-pass (≈ 160 kHz), clamp. A 30 V accident flows into the rail, not the chip.
- **Sampling:** 500 kSPS aggregate; anything above half the sample rate aliases. The Scope tab: rolling tens of kS/s (WiFi-limited), burst up to the ADC.

(c) The op-amp stage as a black box — B3

DAC8563: 0–5 V, 16-bit, 2.5 V reference out

OPA2192 as a **difference amplifier**, 10 kΩ / 40 kΩ:

$$AO = 4\,(V_{DAC} - 2.5\ \mathrm{V}) \;\Rightarrow\; -10\ldots+10\ \mathrm{V}$$

- rails wider than the swing ($\pm$12 V)
- **49.9 Ω** before the SMA: the cable is a capacitor; the scope's 50 Ω input halves what you see
- BAV99 clamp after the resistor
- "clipped" = the output hit its rail — B3's alarm

(d) Isolation — B4

- A **relay**: coil and contacts share no copper. ULN2003 drives four coils from GPIO (the microcontroller's general-purpose pins); its COM pin carries the flyback diodes that eat the coil's kick.
- An **optocoupler** (6N137): LED and transistor share no copper; 5–24 V in, 3.3 V out, fast.
- **Creepage**: the distance that keeps it so — **2.5 mm band**, no copper of any other net (a set of pins connected together) under it, on any layer. DRC (KiCad's design rule check) checks it; your reviewer checks it.
- Contacts: $\leq$ **30 V DC, 1 A. Never mains.**

(e) 3.3 V vs 5 V logic — B5

- ESP32-S3 outputs 0 / 3.3 V and must **never** see 5 V on a pin.
- Lab equipment speaks **5 V TTL**: high $\geq$ 2 V, outputs up to 5 V.
- **74AHCT541**: 3.3 V in $\rightarrow$ 5 V out, eight lines (6 DIO + 2 fast).
- **74LVC1T45**: translates either way under a direction pin $\rightarrow$ the **TRIG** SMA is input *or* output. 100 Ω in series: two drivers on one wire survive.
- TTL into 50 Ω $\approx$ 2.5 V; into 1 MΩ $\approx$ 5 V — the note on the panel.

(f) The buses that tie it together — base

SPI (a common wiring standard for chips to talk to the microcontroller) — three shared wires (SCLK 12, MOSI 11, MISO 13) + one chip-select per device: ADC `/CS` 10, DAC `/SYNC` 5. 33 Ω series on the data lines; short, one direction from the socket.

I²C (the other common one) — two wires (SDA 1, SCL 2), 4.7 kΩ pull-ups, addresses: OLED 0x3C, two Qwiic connectors for anything else.

Every block's driver is "write a few bytes on one of these, read a few back".

Read your block, again

Two harder questions from your block page, with the tutor:

- *Which pins carry the **fastest** signal in your block?*
- *Where does your block's **ground** join the rest — and why only there?*

Then talk to the neighbour who will **review your PR**.
Reviewer ring: **A → B → C → D → E → A**.

KiCad — the only six operations

	Schematic	PCB
1	**Place** `A` — project library `class_board` (our symbols and footprints)	
2	**Wire** `W`, value `V`, fields `E` (footprint!)	
3	**Annotate**, **ERC** (electrical rule check)	
4	*Update PCB from Schematic* `F8` →	footprints land near your zone
5		**Drag** `M`, decoupling next to the pin
6		**Route** `X` (via `V`), **DRC**, 3D `Alt+3`

E6 — schematic gaps

1. `git switch -c b<N>-<name>` (a new branch — your own line of work) • open `hardware/class-board.kicad_pro` • your sheet `student/b<N>_gapped.kicad_sch`
2. Full PDF page for your block beside it. Your block page lists the **3–4 deleted parts**.
3. Each: place → wire → value → **footprint field** → keep the PDF's **reference designator** (the part's label: R12, C3 …)
4. **ERC → 0 errors.**

B1	**one input network (AI4) + the ADC's AVDD decoupling**
B2	one polyfuse + TVS pair (jack path) + 78L05 output caps
B3	AO2 difference-amp resistors + AO2 49.9 Ω / clamp
B4	one relay channel + one opto input network
B5	74HCT125 fast-out parts + one TTL terminal + buffer decoupling

E7 — route your zone (start)

- `F8` → the new footprints appear → drag them **inside** `ZONE_B<N>`, capacitors next to the pin they serve
- Route `X`: F.Cu first, B.Cu for crossings, **nothing on In1.Cu**, widths from the net classes (don't change them)
- **Run DRC early and often.** The two classics: *wrong layer*, *unconnected net*. Read every message.
- Block-specific: B1 nothing digital under the inputs · B2 1.0 mm for +5V, modules ≥ 15 mm from analog · B3 feedback resistors tight to the op-amp · B4 the 2.5 mm band · B5 `FAST` nets short with GND adjacent

Done today: **about half the zone**.

PR + peer review, live

1. `git add hardware/ && git commit && git push -u origin b<N>-<name>` → `gh pr create` — **3D screenshot in the description**
2. *Reviewers* → your ring reviewer • CI must be green

commit = save a snapshot with a message • push = upload it to GitHub • `gh pr create` opens the pull request from the terminal • CI = the automatic build GitHub runs on every pull request, green or red.

3. Reviewer: *Files changed*, check out the branch, open the PCB, walk **the checklist**, ≥ 2 comments, *Approve* / *Request changes*

The checklist (chapter C, the instructor's merge list): ERC 0 / DRC 0 • an `LCSC` part number (the factory's catalogue) on every part • values match the PDF • decoupling next to pins • nothing on In1.Cu • nothing outside the zone • pin 1 marked • ground pour joined • B4 band ≥ 2.5 mm • silkscreen (printed label) on every connector • CI green

Homework 2 — gated

Homework 2 — hand-in dates agreed in class
E7 finish routing — DRC 0, 3D screenshot (V5 first)
E8 one part from LCSC — `easyeda2kicad --full --lcsc_id C…`
E9 PR with the screenshot, request your reviewer
E10 review your peer ($\geq$ 2 comments) • fix yours

Mon 28 Sep: the instructor paste-merges the five zones, full DRC, Gerbers (manufacturing files) / BOM (parts list) / CPL (placement file), quote, **order** — 5 assembled + 2 bare + 6 front panels. Boards back $\approx$ 12 Oct.

Fri 2 Oct: Firmware and Mechanical 101

Your block's driver and phone panel in sim mode (the firmware fakes its hardware, on the dev board) • one alarm rule • your box from the template

No homework the week of 28 Sep. `/tutor HW2` until then.